

作業 1.

1. (1) $|\vec{A}| = \sqrt{1^2+2^2+(-3)^2} = \sqrt{14}$ (2) $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{1}{\sqrt{14}}\hat{x} + \frac{2}{\sqrt{14}}\hat{y} - \frac{3}{\sqrt{14}}\hat{z}$

(3) $\vec{A} \cdot \vec{B} = 1 \times 0 + 2 \times (-4) + (-3) \times 1 = -11$ (4) $\cos \theta_{AB} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = -\frac{11}{\sqrt{14} \cdot \sqrt{17}} = -\frac{11}{\sqrt{238}}$, $\theta_{AB} = \arccos\left(-\frac{11}{\sqrt{238}}\right) = \pi - \arccos\left(\frac{11}{\sqrt{238}}\right)$

(5) $\vec{A} \cdot \hat{C} = \frac{\vec{A} \cdot \vec{C}}{|\vec{C}|} = \frac{1 \times 5 + 2 \times 0 + (-3) \times (-2)}{\sqrt{5^2+2^2}} = \frac{11}{\sqrt{29}}$ (6) $\vec{A} \times \vec{C} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = -4\hat{x} - 13\hat{y} - 10\hat{z}$

(7) $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = -\vec{B} \cdot (\vec{A} \times \vec{C}) = (-4\hat{y} + \hat{z}) \cdot (4\hat{x} + 13\hat{y} + 10\hat{z}) = -52 + 10 = -42$ (8) $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} = 11\vec{B} + 11\vec{C} = 11(5\hat{x} - 4\hat{y} - \hat{z}) = 55\hat{x} - 44\hat{y} - 11\hat{z}$

$\vec{C} \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot (\vec{B} \times \vec{C}) = -42$
 $(\vec{A} \times \vec{B}) \times \vec{C} = -\vec{C} \times (\vec{A} \times \vec{B}) = \vec{C} \times (\vec{B} \times \vec{A})$
 $= (\vec{C} \cdot \vec{A})\vec{B} - (\vec{C} \cdot \vec{B})\vec{A}$
 $= 11\vec{B} + 2\vec{A} = 2\hat{x} - 40\hat{y} + 5\hat{z}$

2. (1) $\nabla \times (\vec{A} \times \vec{B}) = \nabla \times (\vec{A}_c \times \vec{B}) + \nabla \times (\vec{A} \times \vec{B}_c)$
 $= (\nabla \cdot \vec{B})\vec{A}_c - (\vec{A}_c \cdot \nabla)\vec{B} + (\vec{B}_c \cdot \nabla)\vec{A} - (\nabla \vec{A})\vec{B}_c$
 $= (\nabla \cdot \vec{B})\vec{A} - (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} - (\nabla \vec{A})\vec{B}$

(2) $\nabla(\vec{A} \cdot \vec{B}) = \nabla(\vec{A}_c \cdot \vec{B}) + \nabla(\vec{A} \cdot \vec{B}_c)$
 $= (\vec{A}_c \cdot \nabla)\vec{B} + \vec{A}_c \times (\nabla \times \vec{B}) + (\vec{B}_c \cdot \nabla)\vec{A} + \vec{B}_c \times (\nabla \times \vec{A})$
 $= (\vec{A} \cdot \nabla)\vec{B} + \vec{A} \times (\nabla \times \vec{B}) + (\vec{B} \cdot \nabla)\vec{A} + \vec{B} \times (\nabla \times \vec{A})$

(3) 取 (2) 中的 $\vec{B} = \vec{A}$
 $\Rightarrow \nabla A^2 = 2(\vec{A} \cdot \nabla)\vec{A} + 2\vec{A} \times (\nabla \times \vec{A}) \Leftrightarrow \vec{A} \times (\nabla \times \vec{A}) = \frac{1}{2}\nabla A^2 - (\vec{A} \cdot \nabla)\vec{A}$

3. (1) $r = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$

(1.1) $\frac{\partial r}{\partial x} = \frac{2(x-x')}{2\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} = \frac{x-x'}{r}$
同理可得 $\frac{\partial r}{\partial y} = \frac{y-y'}{r}$, $\frac{\partial r}{\partial z} = \frac{z-z'}{r}$
所以 $\nabla r = \frac{x-x'}{r}\hat{x} + \frac{y-y'}{r}\hat{y} + \frac{z-z'}{r}\hat{z} = \frac{\vec{r}}{r}$
 $\frac{\partial r}{\partial x'} = \frac{-2(x-x')}{2\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} = -\frac{x-x'}{r} = -\frac{\partial r}{\partial x}$
所以 $\nabla' r = \frac{\partial r}{\partial x'}\hat{x} + \frac{\partial r}{\partial y'}\hat{y} + \frac{\partial r}{\partial z'}\hat{z} = -\nabla r = -\frac{\vec{r}}{r}$
即 $\nabla r = -\nabla' r = \frac{\vec{r}}{r}$

(1.3) $\nabla \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-x' & y-y' & z-z' \end{vmatrix} = \sum_{cyc} \left(\frac{\partial(z-z')}{\partial y} - \frac{\partial(y-y')}{\partial z} \right) \hat{x} = 0$
 $\nabla' \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x'} & \frac{\partial}{\partial y'} & \frac{\partial}{\partial z'} \\ x-x' & y-y' & z-z' \end{vmatrix} = \sum_{cyc} \left(\frac{\partial(z-z')}{\partial y'} - \frac{\partial(y-y')}{\partial z'} \right) \hat{x} = 0$
 $\Rightarrow \nabla \times \vec{r} = -\nabla' \times \vec{r} = 0$

(1.2) $\nabla \cdot \vec{r} = \frac{\partial(x-x')}{\partial x} + \frac{\partial(y-y')}{\partial y} + \frac{\partial(z-z')}{\partial z} = 1+1+1=3$
 $\nabla' \cdot \vec{r} = \frac{\partial(x-x')}{\partial x'} + \frac{\partial(y-y')}{\partial y'} + \frac{\partial(z-z')}{\partial z'} = -1-1-1=-3$
 $\Rightarrow \nabla \cdot \vec{r} = -\nabla' \cdot \vec{r} = 3$

(1.4) $\nabla \frac{1}{r} = \frac{r(\nabla 1) - 1 \nabla r}{r^2} = -\frac{\nabla r}{r^2} = -\frac{\vec{r}}{r^3}$
 $\nabla' \frac{1}{r} = \frac{r(\nabla' 1) - 1 \nabla' r}{r^2} = \frac{\vec{r}}{r^3}$
即 $\nabla \frac{1}{r} = -\nabla' \frac{1}{r} = -\frac{\vec{r}}{r^3}$

(1.5) $\nabla \cdot \frac{\vec{r}}{r^3} = \nabla \left(\frac{1}{r^3} \right) \cdot \vec{r} + \frac{1}{r^3} \nabla \cdot \vec{r}$
 $= -\frac{\nabla r^3}{r^6} \cdot \vec{r} + \frac{3}{r^3} = -\frac{3r^2 \nabla r}{r^6} \cdot \vec{r} + \frac{3}{r^3}$
 $= -\frac{3}{r^4} \cdot \frac{r^2}{r} + \frac{3}{r^3} = 0 \quad (r \neq 0)$
 $\nabla' \cdot \frac{\vec{r}}{r^3} = \nabla' \left(\frac{1}{r^3} \right) \cdot \vec{r} + \frac{1}{r^3} \nabla' \cdot \vec{r}$
 $= -\frac{\nabla' r^3}{r^6} \cdot \vec{r} - \frac{3}{r^3} = -\frac{3r^2 \nabla' r}{r^6} \cdot \vec{r} - \frac{3}{r^3}$
 $= \frac{3}{r^4} \cdot \frac{r^2}{r} - \frac{3}{r^3} = 0 \quad (r \neq 0)$
即 $\nabla \cdot \frac{\vec{r}}{r^3} = -\nabla' \cdot \frac{\vec{r}}{r^3} = 0 \quad (r \neq 0)$

(1.6) $\nabla \times \frac{\vec{r}}{r^3} = \nabla \times \nabla \left(-\frac{1}{r^3} \right) = 0$

(2) 设 $\vec{a} = a_x \hat{x} + a_y \hat{y} + a_z \hat{z} = \sum_{j=1}^3 a_j \hat{e}_j$
 $\vec{a} \cdot \nabla = \sum_{j=1}^3 a_j \frac{\partial}{\partial x_j}$

(2.1) $(\vec{a} \cdot \nabla) \vec{r} = \sum_{j=1}^3 a_j x \frac{\partial \vec{r}}{\partial x_j} = \sum_{j=1}^3 a_j x \hat{e}_j = \vec{a}$ (2.2) $\nabla(\vec{a} \cdot \vec{r}) = (\vec{r} \cdot \nabla) \vec{a} + (\vec{a} \cdot \nabla) \vec{r} + \vec{r} \times (\nabla \times \vec{a}) + \vec{a} \times (\nabla \times \vec{r})$
 $= 0 + \vec{a} + 0 + 0 = \vec{a}$

(2.3) $\nabla \cdot (\vec{E}_0 \sin(\vec{k} \cdot \vec{r})) = \sin(\vec{k} \cdot \vec{r}) \nabla \cdot \vec{E}_0 + \nabla(\sin(\vec{k} \cdot \vec{r})) \cdot \vec{E}_0$ (2.4) $\nabla \times (\vec{E}_0 \sin(\vec{k} \cdot \vec{r})) = \sin(\vec{k} \cdot \vec{r}) \nabla \times \vec{E}_0 + \nabla \sin(\vec{k} \cdot \vec{r}) \times \vec{E}_0$
 $= \cos(\vec{k} \cdot \vec{r}) \nabla(\vec{k} \cdot \vec{r}) \cdot \vec{E}_0$
 $= \cos(\vec{k} \cdot \vec{r}) (\vec{k} \cdot \vec{E}_0)$
 $= \cos(\vec{k} \cdot \vec{r}) \vec{k} \times \vec{E}_0$